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[18MTAPT201]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

HEAT TRANSFER AND COMBUSTION

Time: 03 Hours **Maximum Marks:100**

Instructions: Answer five full questions. Question numbers one and two are compulsory

1. a. A copper slab (k=372 W/m-K) is 4 mm thick. It is protected from corrosion on each side by a 2mm thick layer of stainless steel (k=18 W/m-K). The temperature is 400 ^{0}C on one side of this composite wall and 90 ^{0}C on the other. Find the temperature distribution in the copper slab and heat conducted through the wall. **06**

   b. Derive an expression for the variation of temperature with time of a body placed in a fluid stream using lumped system analysis. Consider the case of cooling. State the assumption made. **06**

   c. A thermocouple used to measure the temperature of hot air flowing in a duct whose walls are maintained at T_w=410K, shows a temperature reading of T_{th}=640K, refer Fig.1c. Assuming emissivity of thermocouple junction to be 0.6 and the convection heat transfer coefficient to be h= 90 W/m^2-K, determine the actual temperature of the air, T_f. **08**

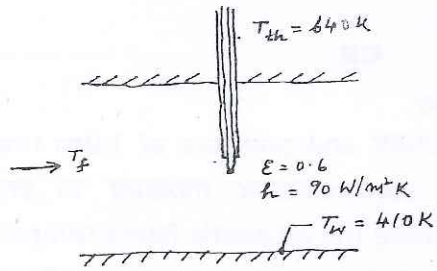

Fig.1c

2. a. Given the eularian velocity vector field $\vec{V} = 3t\vec{i} + xy\vec{j} + ty^2\vec{k}$ find the total acceleration of fluid particle. **08**

   b. In cartesian coordinate system, Newton's second law of motion can be stated as $F_x=ma_x$, F_x, is the sum of x-component of all forces acting on the fluid element. Apply this law to a moving fluid element in a viscous flowfield and develop the x-component of 3D momentum equation of the general compressible fluid flow, in Cartesian co-ordinate system. Draw a sketch indicating the various x-component forces acting on the fluid element. **12**

3. a. Write a note of scales of turbulence **10**

   b. Air at 30^{0}C and atmospheric pressure is flowing over a flat plate at a velocity of 5m/s. If the plate is 30cm wide and at a temperature of 70^{0}C, calculate the following quantities at x=0.3m: **10**

   i. Thickness of velocity and thermal boundary layers

   ii. Local and average skin friction coefficient

   iii. Local and average heat transfer coefficient

   iv. Total drag on the plate.

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Take the following properties of air at 323K: $\rho=1.09$ kg/m^3; $\nu=18X10^{-6}$ m^2/s; k=0.0273 W/m ^{0}C; C_p=1.007 KJ/kg ^{0}C; Pr=0.723

Use the correlation $Nu_x=0.332\ Re_x^{0.5}\ Pr^{1/3}$

OR

4. a. Explain the concept of velocity and thermal boundary layers over a flat plate with diagrams. Draw a schematic diagram of growth of velocity boundary with respect to distance from leading edge in the case of flow of air over a flat plate. Indicate the variation of local skin friction coefficient in the diagram. **10**

b. Consider the heating of fluid in a tube of constant cross section whose inner surface is maintained at constant temperature of T_s. The mean temperature of the fluid T_m increases in the flow direction as a result of heat transfer. Derive the expression for logarithmic mean temperature difference. **10**

5. a. Discuss some of the film cooling devices in combustion chamber liner with schematic diagrams. **07**

b. A natural gas-fired industrial boiler operates with an oxygen concentration of 3 mole percent in the flue gases. Determine the operating air-fuel ratio and the equivalence ratio. Treat the natural gas as methane. **07**

c. Distinguish between elementary and global (overall) reaction in chemical kinetics with example for each. **06**

OR

6. a. Write a note on turbine blade cooling. **07**

b. In a burner, hydrogen and air at 298K and pressure of 1atm undergo constant pressure combustion process for stochiometric mixture to produce product mixture of water and nitrogen. Estimate the adiabatic flame temperature, T_{ad}. The product mixture enthalpy is estimated using constant specific heat evaluated around 1350 K ($\approx 0.5(298+T_{ad})$). **07**

Use the following data:

Species	Enthalpy of formation @298 K (kJ/kmol)	Specific heat @ 1350 K (kJ/kmol-K)
H_2O	-241826	45.49
N_2	--	34.34

c. Define laminar burning velocity. Discuss the variation of burning velocity with equivalence ratio for typical hydrocarbon fuel- air and CO-air mixtures **06**

7. a. Explain about the stability of the laminar flame front in the Bunsen burner. **08**

b. How is Damköhler number expressed and what is its significance? **05**

c. Write a note on combustion instability in combustion chamber. **07**

OR

8. a. Explain structure of turbulent premixed flame stabilized at the rim of a Bunsen burner with diagrams. **10**

b. Explain Rayleigh's criterion for combustion instability **10**